

LANAI, HI, USA

WMO: 911905

Lat: 20.790N

Lon: 156.949W

Elev: 396

StdP: 96.66

Time Zone: -10.00 (W10)

Period: 05-19

WBAN: 22524

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	14.8	15.5	10.7	8.4	17.7	11.7	9.0	18.2	16.1	20.0	14.7	19.0	2.2	80	0.382

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.7	28.8	22.0	28.1	21.9	27.7	21.7	24.1	26.1	23.5	25.8	23.1	25.5	6.4	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.5	19.3	24.8	22.9	18.5	24.4	22.4	17.9	24.2	74.6	26.2	72.3	25.8	70.5	25.7	26.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
12.8	11.5	10.5	DB	12.7	29.6	0.6	0.5	12.3	30.0	12.0	30.3	11.7	30.5	11.2	30.9
			WB	11.2	24.4	0.6	1.1	10.8	25.2	10.4	25.8	10.1	26.4	9.7	27.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	22.0	20.2	20.3	20.4	21.4	21.8	22.8	23.6	23.8	23.7	23.2	22.1	21.0
	DBStd	1.69	1.14	1.35	1.08	1.02	1.03	0.80	0.89	1.01	1.08	1.04	1.05	1.14
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	2	1	1	1	0	0	0	0	0	0	0	0	0
	CDD10.0	4396	317	288	323	342	367	383	421	429	412	409	362	342
	CDD18.3	1356	59	56	66	92	109	133	163	171	162	151	112	83
	CDH23.3	5008	111	106	99	209	342	534	754	891	818	662	323	158
	CDH26.7	395	3	4	1	3	12	27	64	108	94	61	13	4
(14) Wind	WSAvg	4.4	3.9	4.5	4.6	5.1	4.1	4.6	4.9	4.8	4.3	3.9	4.2	4.3
(15) Precipitation	PrecAvg	952	151	101	95	78	81	35	50	38	55	71	95	118
	PrecMax	1688	429	347	324	364	276	164	189	105	225	291	420	264
	PrecMin	433	11	14	4	7	8	1	6	4	6	2	4	5
	PrecStd	318	126	87	77	80	74	36	41	26	55	65	97	82
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.1	27.2	26.9	27.2	27.9	28.3	29.1	29.5	29.6	29.3	28.1	27.4
		MCWB	20.1	20.6	20.4	20.5	21.3	21.3	22.3	21.9	22.6	22.3	21.0	20.9
	2%	DB	25.6	25.8	25.4	26.1	26.9	27.5	28.1	28.6	28.5	28.1	26.9	25.9
		MCWB	19.7	20.0	19.7	20.2	21.0	21.1	21.9	22.0	22.2	22.0	20.7	20.2
	5%	DB	24.5	24.7	24.6	25.3	26.1	26.8	27.5	27.9	27.8	27.4	26.0	25.0
		MCWB	19.3	19.5	19.4	19.9	20.7	21.0	21.8	21.9	22.1	21.7	20.5	20.0
	10%	DB	23.7	23.7	23.8	24.6	25.2	26.1	26.8	27.2	27.1	26.6	25.2	24.1
		MCWB	19.0	19.1	19.2	19.7	20.3	20.9	21.7	21.8	22.0	21.5	20.2	19.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.6	22.4	21.4	22.1	22.8	23.6	24.3	25.0	24.9	24.5	22.7	22.8
		MCDB	23.7	24.0	24.8	24.2	25.0	25.0	26.0	26.5	26.5	26.2	25.3	24.5
	2%	WB	20.8	21.5	20.8	21.3	22.1	22.6	23.5	24.0	24.0	23.4	21.9	21.7
		MCDB	23.1	23.5	23.5	23.9	24.9	25.7	26.2	26.2	26.2	25.8	24.5	23.6
	5%	WB	20.4	20.7	20.4	20.7	21.5	21.9	23.0	23.3	23.4	22.8	21.4	21.1
		MCDB	22.9	22.9	23.1	23.8	24.5	25.1	25.5	25.7	25.8	25.6	24.2	23.2
	10%	WB	19.9	20.1	19.9	20.3	20.9	21.4	22.5	22.8	22.9	22.3	20.9	20.6
		MCDB	22.7	22.5	22.7	23.4	24.1	24.8	25.3	25.4	25.5	25.2	23.9	22.8
(35) Mean Daily Temperature Range	5% DB	MDBR	7.9	7.6	7.2	7.4	7.8	7.7	7.5	7.7	7.6	7.7	7.3	7.1
		MCDBR	9.0	9.0	8.2	8.0	8.4	8.4	7.9	8.2	8.1	8.5	8.1	8.1
	5% WB	MCWBR	4.6	4.5	4.1	3.7	3.9	3.6	3.5	3.6	3.6	3.8	3.8	4.1
		MCWBR	7.7	7.5	7.0	7.4	7.8	7.5	6.9	6.6	6.9	7.5	7.1	7.0
(40) Clear-Sky Solar Irradiance	taub		0.333	0.338	0.347	0.362	0.359	0.347	0.346	0.346	0.339	0.346	0.343	0.331
		taud	2.653	2.631	2.614	2.542	2.581	2.624	2.639	2.642	2.668	2.655	2.662	2.680
	Edn at Noon		948	960	959	940	932	935	937	943	953	939	930	940
		Edh at Noon	86	92	97	106	101	96	95	95	91	89	84	81
(44) All-Sky Solar Radiation	RadAvg		4.55	5.12	5.76	6.55	6.95	7.19	7.11	6.89	6.44	5.46	4.62	4.17
	RadStd		0.20	0.33	0.55	0.38	0.37	0.19	0.15	0.27	0.18	0.15	0.11	0.34

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	+90	+91

Nomenclature: See separate page